SE LAB 7 - FUZZING FOR INPUT VALIDATION BUGS

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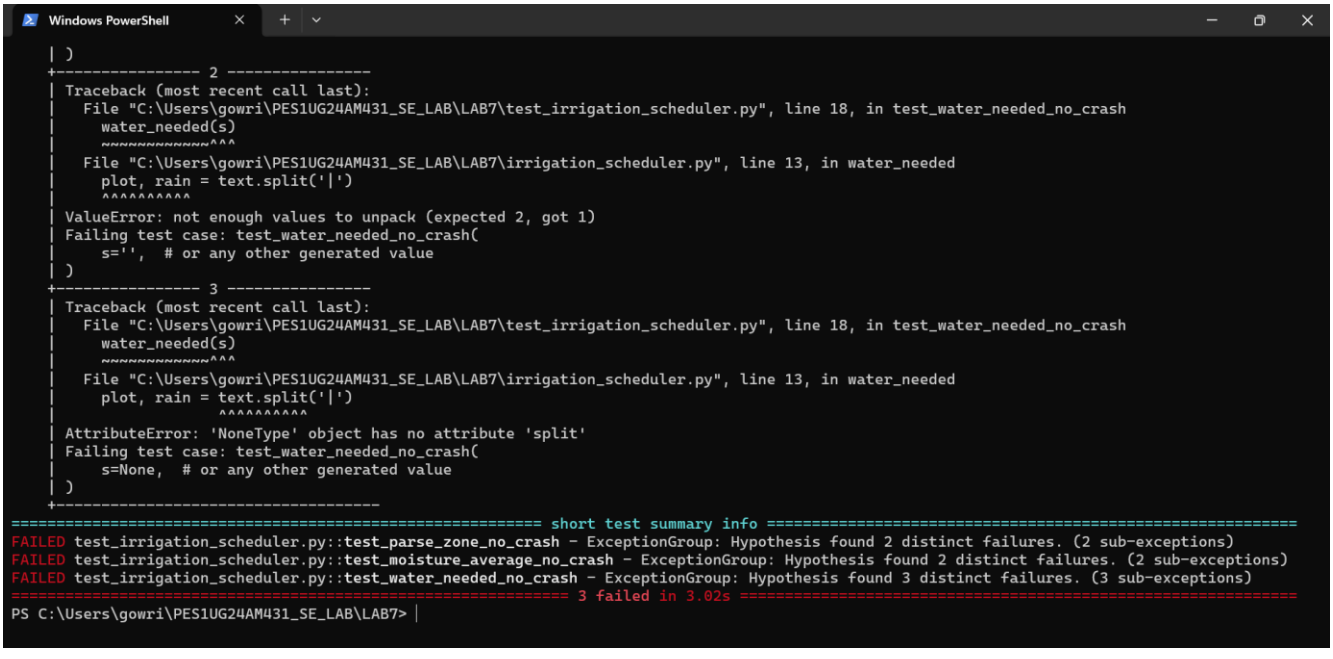
PES1UG24AM431

OBJECTIVE - To use property-based fuzz testing with Hypothesis and pytest to identify input-validation bugs in the irrigation scheduler application, improve the input validation, and verify the corrected program using automated tests.

Application Functions Tested

Function	Purpose
parse_zone(text)	Parses a zone name and area from zone:area input.
moisture_average(readings_csv)	Calculates the average of comma-separated moisture readings.
water_needed(text)	Calculates the water required from area|rainfall input.

Evidence – Initial Fuzz Testing Failures



Evidence – Fuzz Tests After Input Validation Fixes

```
PS C:\Users\gowri\PES1UG24AM431_SE_LAB\LAB7> py -m pytest -v test_irrigation_scheduler.py
====================== test session starts ======================
platform win32 -- Python 3.14.2, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\gowri\AppData\Local\Python\pythoncore-3.14-64\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\gowri\PES1UG24AM431_SE_LAB\LAB7
plugins: hypothesis-6.170.15
collected 3 items

test_irrigation_scheduler.py::test_parse_zone_no_unexpected_crash PASSED [ 33%]
test_irrigation_scheduler.py::test_moisture_average_no_unexpected_crash PASSED [ 66%]
test_irrigation_scheduler.py::test_water_needed_no_unexpected_crash PASSED [100%]

======================= 3 passed in 0.62s =======================
PS C:\Users\gowri\PES1UG24AM431_SE_LAB\LAB7>
```

Evidence – Valid Application Execution

The corrected application was executed with valid sample inputs:

- Zone input: north:120
- Moisture readings: 40,35,50
- Area and rainfall: 120|8

The expected outputs were:

- Zone and Area: ('North', 120.0)
- Average Moisture: 41
- Water Needed: 592.0

```
PS C:\Users\gowri\PES1UG24AM431_SE_LAB\LAB7> py irrigation_scheduler.py
==== Running parse_zone ====
Enter zone:area in m2 (e.g. north:120): north:120
Zone and Area: ('North', 120.0)

==== Running moisture_average ====
Enter moisture readings as CSV (e.g. 40,35,50): 40,35,50
Average Moisture: 41

==== Running water_needed ====
Enter area|rainfall (e.g. 120|8): 120|8
Water Needed: 592.0
PS C:\Users\gowri\PES1UG24AM431_SE_LAB\LAB7>
```

Result

The irrigation scheduler was tested using Hypothesis-based fuzz testing. The initial tests revealed input-validation failures. After adding validation and updating the tests, all three tests passed, and the application produced the expected results for the valid sample inputs.